

0/17 Questions Answered

hw6-single-prop-gluten-free-pasta

Student Name

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Q1**6 Points**

Note: There are three options for homework 6, that differ by data topic, and you only need to complete one of the three options.

Celiac disease, non-celiac gluten sensitivity, and wheat allergy are three forms of gluten intolerance. Gluten intolerance is on the rise in the US, so there is interest in finding enjoyable and nutritious gluten-free alternatives.

Consider the following scenario:

Maria, a student in public health science, is conducting a small research study on gluten-free alternatives.

She obtains a random sample of 20 UCI students and asks them whether they prefer gluten-free alternatives. The results show that 10% of the students in her sample prefer gluten-free alternatives. She wants to construct a 95% confidence interval for the proportion of people who prefer gluten-free alternatives, among all UCI students, so she computes the margin of error, which turns out to be 0.1315.

Q1.1**2 Points**

What is the upper bound of this CI? Enter only your final answer. Round to four decimal places (1.23456 should be rounded to 1.2346). Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

Q1.2**1 Point**

What is the lower bound of this CI? Enter only your final answer. Round to four decimal places (1.23456 should be rounded to 1.2346). Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q1.3****3 Points**

Maria would like to include her results in an article that she is writing for a local newspaper but, when she computes the CI, she feels puzzled: something doesn't seem right about the lower bound. She knows you have taken a stats class, and she comes to you to ask your opinion. What is the issue with the lower bound of her CI? And what is a reason that you might suggest to Maria to explain why she faced this issue?

Save Answer

Q2 Breadfruit Flour Pasta

11 Points

Breadfruit is a nutritious fruit which can be processed into a flour, providing an inexpensive gluten-free flour alternative in tropical areas where the fruit is grown. A study conducted by Nochera and Ragone (2019) investigated the creation and evaluation of a gluten-free breadfruit flour pasta. Untrained panelists did a sensory evaluation of the breadfruit flour pasta and wanted to investigate if a majority of people like the pasta. 57 of the 71 panelists indicated that they liked the pasta.

Q2.1

1 Point

First we want to understand what are we investigating and what is our data?

The population parameter of interest for this study is:

Proportion of panelists that like the breadfruit flour pasta

Proportion of people that will like the breadfruit flour pasta

Number of panelists that like the breadfruit flour pasta

Number of people that will like the breadfruit flour pasta

Save Answer

Q2.2**1 Point**

The value of the population parameter is:

57

71

 $57/71 = 0.80$ $71 - 57 = 14$

unknown

[Save Answer](#)**Q2.3****1 Point**

The point estimate (sample statistic) for this study is:

Proportion of panelists that like the breadfruit flour pasta

Proportion of people that will like the breadfruit flour pasta

Number of panelists that like the breadfruit flour pasta

Number of people that will like the breadfruit flour pasta

[Save Answer](#)

Q2.4

1 Point

The value of the point estimate is:

57

71

 $57/71 = 0.80$ $71 - 57 = 14$

unknown

[Save Answer](#)

Q2.5

1 Point

What are the null and alternative hypotheses to test if a majority (more than 50%) of people will like the breadfruit flour pasta?

H_0 : The proportion of people that will like the breadfruit flour pasta is more than 50%; H_1 : The proportion of people that will like the breadfruit flour pasta is less than 50%

H_0 : The proportion of people that will like the breadfruit flour pasta is 50%; H_1 : The proportion of people that will like the breadfruit flour pasta is less than 50%

H_0 : The proportion of people that will like the breadfruit flour pasta is 50%; H_1 : The proportion of people that will like the breadfruit flour pasta is not 50%

H_0 : The proportion of people that will like the breadfruit flour pasta is not 50%; H_1 : The proportion of people that will like the breadfruit flour pasta is 50%

[Save Answer](#)

Q2.6**1 Point**

Let's check that the conditions are met to do a hypothesis test for a proportion. For each condition specify if it is met and explain your answer.

Condition 1: Sample data are independent?

Yes

No or unknown

Save Answer

Q2.7**1 Point**

Condition 2: More than 10 success and 10 failures in the sample?
Explain your answer.

Yes

No

Save Answer

Q2.8**1 Point**

Condition 3: Sample size is less than 10% of population? Explain your answer.

Yes

No

Save Answer

Q2.9**1 Point**

We will proceed assuming all conditions for the hypothesis test were met. If a one-sample test for proportion is performed, which of the following outcomes would lead to a rejection of the null hypothesis at a significance level of 0.05?

A p-value of 0.07

A p-value of 0.03

A p-value of 0.50

A p-value of 1.00

Save Answer

Q2.10**1 Point**

Conducting a hypothesis test for one proportion results in a p-value of 0.0015. Which of the following is a valid conclusion from this p-value?

We accept the null hypothesis and conclude that the population proportion of people who will like the breadfruit flour pasta is 50%.

We reject the null hypothesis and conclude that the population proportion of people who will like the breadfruit flour pasta is 50%.

We reject the null hypothesis and conclude the population proportion of people who will like the breadfruit flour pasta is not 50%.

We fail to reject the null hypothesis and cannot make a conclusion about the population proportion of people who will like the breadfruit flour pasta.

[Save Answer](#)

Q2.11**1 Point**

In this study they actually wanted to approximate the true proportion of people that like the breadfruit flour pasta, so they computed a 95% confidence interval and got (71%, 89.5%). What is an appropriate interpretation of this confidence interval?

The true proportion of people who will like the breadfruit flour pasta is somewhere between 71% and 89.5%.

We are confident that the true proportion of people who will like the breadfruit flour pasta is somewhere between 71% and 89.5%.

We are 95% confident that the true proportion of people who will like the breadfruit flour pasta is somewhere between 71% and 89.5%.

We are confident that there is a 95% probability that the true proportion of people who will like the breadfruit flour pasta is somewhere between 71% and 89.5%.

There is a 95% probability that the true proportion of people who will like the breadfruit flour pasta is somewhere between 71% and 89.5%.

Save Answer

Q3**1 Point**

The following two questions are related to the homework choice you made in choosing to do this assignment instead of the other two option. There are no correct answers, this question is only being graded for completeness (note that answering "I felt like it" is an insufficient response).

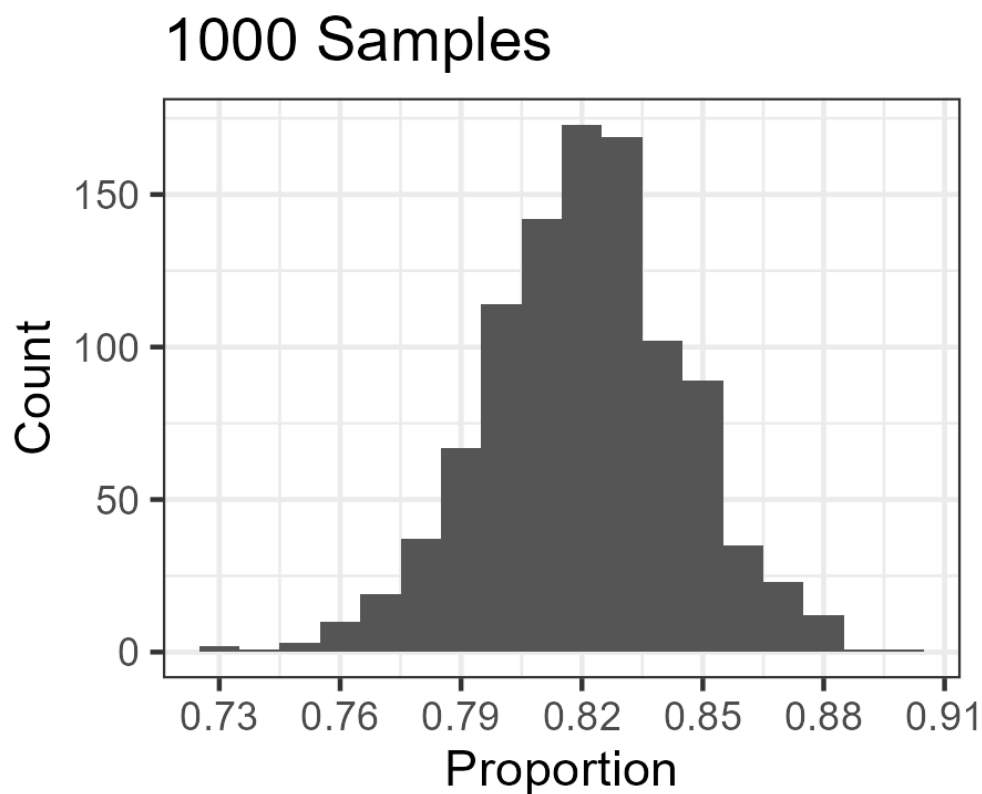
Please explain in detail what led you to choose this topic, gluten-free pasta, to investigate.

What was it about the other two options (cheating with AI and home-team advantage) that led you not to choose them? Please explain in detail.

[Save Answer](#)

Q4**3 Points**

Say we were able to collect 1000 samples, and for each we calculated the sample proportion, plotted below.

**Q4.1****1 Point**

What is visualized in the plot?

- sampling distribution
- distribution of sample data
- distribution of population data

[Save Answer](#)

Q4.2**2 Points**

Based on this plot, what do you think the value of the true population proportion is? Explain.

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